# cloudtorch / develop — Step 5a outline

Explicit regularisation for the *given-background* PINN (step 4a). A short **delta against step 5**: because the background is fixed (0 DoF), the entire *background-control* half of step 5 disappears — only the **cloud** smoothness and bounds remain.

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### 1. OBJECTIVE (THE DELTA)

Add step 5's explicit penalties to the step-4a loss, but on the cloud fields *only*:

$$\mathcal{L} = \chi^2 \; + \; \mathcal{R}_{\text{smooth}} \; + \; \mathcal{R}_{\text{bound}} \qquad \text{(no background terms).}$$

Step 5 existed to *tame a degenerate background*; step 4a has none, so 5a is not about the background at all — it is about giving the **8 free cloud fields** explicit space/time smoothness and physical hinges on top of the implicit Fourier smoothness.

### 2. CARRIED OVER FROM STEP 5 (REUSE `REGULARIZERS.PY` VERBATIM)

The two penalties are *unchanged*, just applied to the cloud groups:

- $\mathcal{R}_{\text{smooth}} = \sum_k \big[ w_k^{xy} \langle (\partial_x p_k)^2 + (\partial_y p_k)^2 \rangle + w_k^t \langle (\partial_t p_k)^2 \rangle \big]$, autograd coordinate-derivatives (`create_graph=True`), mesh-free $\Rightarrow$ works with random minibatches; one extra backward through the *MLP* only.
- $\mathcal{R}_{\text{bound}} = \sum_k \big[ w_k^{\ell} \langle \mathrm{ReLU}(\ell_k - p_k)^2 \rangle + w_k^{u} \langle \mathrm{ReLU}(p_k - u_k)^2 \rangle \big]$, soft hinges per quantity. As in step 5, the step-4 *strict* transforms relax to positivity-only (softplus on $S, \Delta\tau, \Delta v$); $v_{\text{los}}$ goes free, its limit owned by a hinge.

### 3. WHAT DROPS (WHY 5A IS SHORTER)

No background means the whole "control the background" machinery of step 5 is *gone*: no strong smoothing of the 6 PCA coefficients, **no** `no_emission` spectrum cap, **no** background *anchor*. The emission-spike/core-fill degeneracy those fought *cannot arise* (0 background DoF), so 5a keeps only the cloud groups of the `REG` config.

### 4. THE POINT: TAME THE CLOUD PATHOLOGIES 4A EXPOSED

Target exactly what the step-4a maps showed: cap $|v_{\text{los}}| \le v_{\max}$ and $\Delta v$ (hinges) to remove the runaway tails (a few pixels drove $v_{\text{los},1}$ to $-128$, $\Delta v_1$ to 68 reaching for off-core structure), and apply **light** space+time smoothing to the cloud maps — most useful on the weakly-constrained $v_{\text{los},1}$. Hinges are simply the *soft* counterpart of 4a's hard `freeze`. Off-core fidelity stays background-limited (unchanged — clouds own the core).

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Plan only — no code until you approve. Then the same `regularizers.py` + a `REG` dict with cloud groups only (`n_bkg`=0), and a short weight scan in the step-4a notebook. Wiring `background_mode`+`REG` through `fit_cube` is the step-6 integration.